

```
from google.colab import files
uploaded = files.upload()
```

Screenshot ... 142020.png

Screenshot 2026-07-13 142020.png(image/png) - 268920 bytes, last modified: 13/07/2026 - 100% done
Saving Screenshot 2026-07-13 142020.png to Screenshot 2026-07-13 142020 (2).png

```
import cv2
import matplotlib.pyplot as plt

# Try loading the image without forcing grayscale first
img_color = cv2.imread('/content/Screenshot 2026-07-13 142020 (2).png')

if img_color is None:
    print("Error: Could not load image from the specified path.")
else:
    # Convert to grayscale if it's a color image
    if len(img_color.shape) == 3:
        img = cv2.cvtColor(img_color, cv2.COLOR_BGR2GRAY)
    else: # Already grayscale or 1-channel
        img = img_color

    sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
    sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

    sobel_xy = cv2.magnitude(sobel_x, sobel_y)

    plt.imshow(sobel_xy, cmap='gray')
    plt.title('Sobel Edge Detection - XY Axis')
    plt.axis('off')
    plt.show()
```

Sobel Edge Detection - XY Axis

